Firm Entry and AI:

Evidence from US Industries

Guillermo Gallacher*

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Abstract

In this short paper, I show that, following the release of ChatGPT in November 2022, firm entry increased in AI-exposed sectors, temporarily in early 2023 and then more persistently from 2024 onwards. Using monthly sector-level business application data from the Census Bureau Business Formation Statistics, I find that the effect is [X]% larger for a sector at the 75th versus 25th percentile of AI exposure.

*Email: guillermogallacher@gmail.com. Web: guillgall.github.io.

1 Introduction

After the COVID-19 shock, aggregate business applications surged (Decker and Haltiwanger, 2023). The release of ChatGPT in November-2022 seems to have increased firm entry, too, specially after 2024, as seen in Figure 1. This expansion is not uniform across sectors.

Sectors such as Professional Services have increased more than the average, while others such as Construction have lagged behind. From this plot a simple observation arises: while the former sector is exposed to AI, the latter ones is less so. Does this pattern hold across all sectors, though?

In this paper I show that it does: AI-exposed sectors have experienced a firm entry boom, while less-exposed sectors have lagged behind. The scatter-plot in Figure 2 illustrates this clearly.

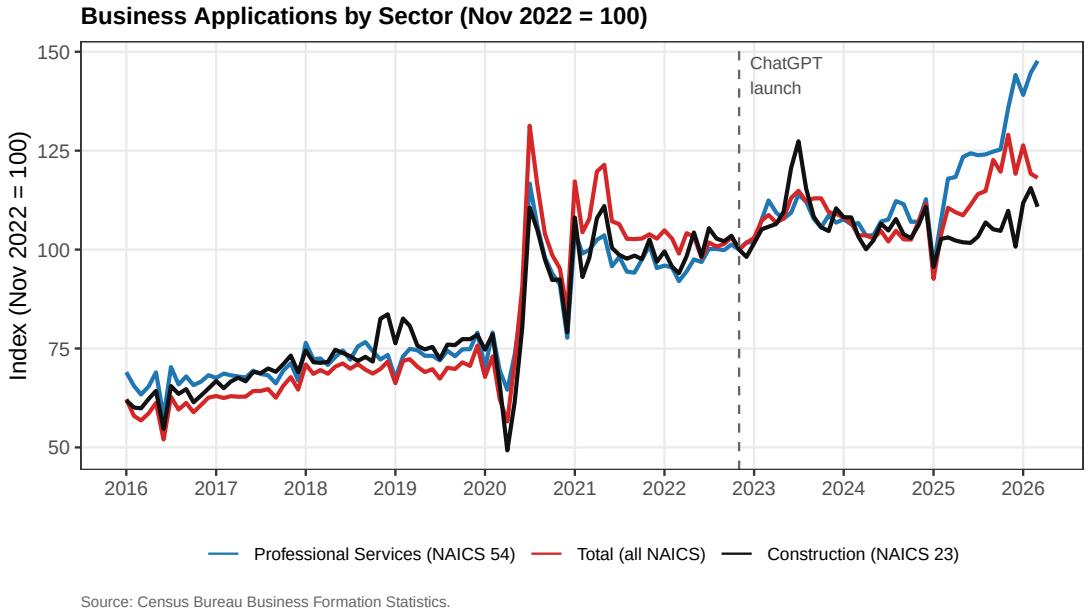

Figure 1: Business applications indexed to November 2022 = 100. Professional Services (NAICS 54), Total (all NAICS), and Construction (NAICS 23). Seasonally adjusted.

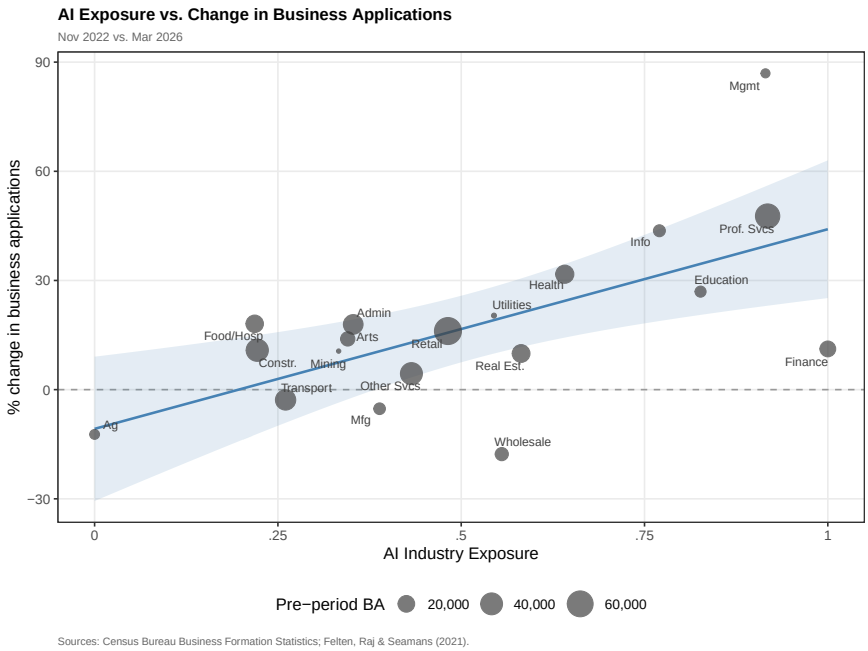

Figure 2: AI exposure versus change in business applications by 2-digit NAICS sector.

I also study the timing of this relationship: I show that it held temporarily in early 2023, but then more persistently from 2024 onwards.

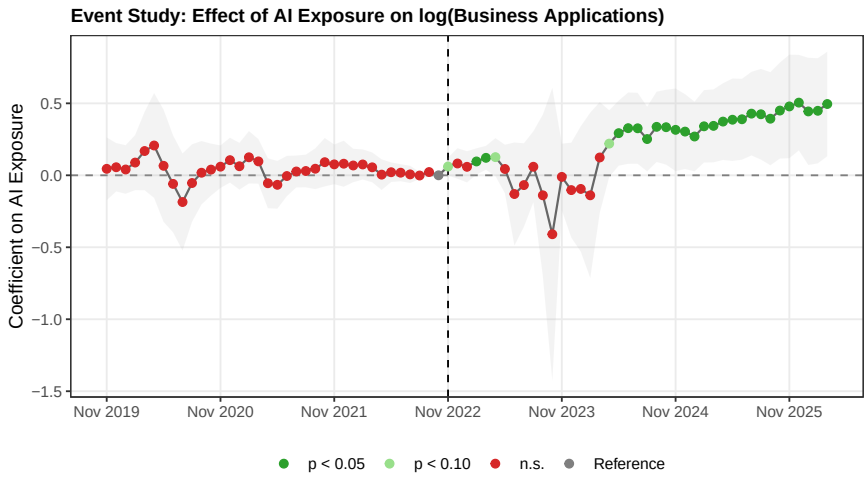

Figure 3: Event-study estimates: effect of AI exposure on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on sector-clustered standard errors.

Contribution to the literature Most of the "economics of AI" debate focuses on workers: which jobs are at risk, and for whom. But a parallel question may be equally consequential:

does AI alter the incentives to start a business?

Chatterji et al. (2025) provide a conceptual treatment of how transformative AI reshapes firm organisation, adoption incentives, and competitive dynamics—a natural complement to the reduced-form evidence presented here. Felten et al. (2021) construct the AIOE/AIIE index by combining O*NET task measures with AI capability scores.

I combine the AI literature with the firm entry dimension, by using business formation data as in Decker and Haltiwanger (2023), and papers cited therein.

2 Data

2.1 Business Formation Statistics

The Census Bureau BFS provides monthly, seasonally adjusted counts of new Employer Identification Number (EIN) applications at the 2-digit NAICS level, derived from IRS Form SS-4 filings. EIN applications are a leading indicator of actual business formation (Decker and Haltiwanger, 2023).

2.2 AI Exposure

Following Felten et al. (2021), we use the AI Industry Exposure (AIIE) index, which maps occupational AI exposure scores to 4-digit NAICS industries through BLS employment weights. We aggregate to 2-digit NAICS using employment-weighted averaging, with 2019 BLS OES industry employment counts as weights, consistent with Felten et al.'s own use of 2019 employment in constructing the index.[1] Table 6 in Appendix A reports the AIIE score alongside summary statistics of the BFS series for each sector in the sample.

[1] Results are unchanged when using an unweighted mean across 4-digit sub-industries.

3 Results

3.1 Regression

Figure 2 plots the percentage change in business applications from November 2022 to the most recent month available against each sector's AI exposure.

Table 1 formalises the scatter with a cross-sectional regression of the form

$$\Delta\overline{\mathrm{BA}}_i \;=\; \alpha + \delta\,\mathrm{AIIE}_i + \varepsilon_i, \tag{1}$$

where $\Delta\overline{\mathrm{BA}}_i$ is the percentage change in business applications from November 2022 to the most recent month available for sector i, and AIIE_i is the normalised AI Industry Exposure index. Column (1) is unweighted OLS; column (2) weights observations by pre-period mean business applications, giving larger sectors proportionally more influence. The slope is positive and statistically significant in both specifications, and the weighted estimate is somewhat larger, indicating that the positive relationship between AI exposure and application growth is stronger among economically larger sectors.

3.2 Difference-in-Differences

I estimate the effect of AI exposure (AIIE_i) on business applicacations (BA_{it}) using a difference-in-differences (DiD) model of the form

$$\log(\mathrm{BA}_{it}) \;=\; \alpha_i + \gamma_t + \beta\,(\mathrm{Post}_t \times \mathrm{AIIE}_i) + \varepsilon_{it}, \tag{2}$$

where i indexes 2-digit NAICS sectors, t indexes calendar months, Post_t is an indicator equal to one from November 2022 onwards, α_i are sector fixed effects absorbing time-invariant differences in application rates, γ_t are month fixed effects absorbing aggregate cyclical variation, and standard errors are clustered by sector. The coefficient β identifies whether sectors with higher AI exposure experienced a differential change in business formation after ChatGPT.

Table 1: Cross-Sectional Regression: AI Exposure and Change in Business Applications

	OLS	WLS
AI Exposure (AIOE, normalised)	54.856***	45.673***
	(16.196)	(12.561)
Constant	-10.773	-5.947
	(9.400)	(7.085)
Sectors	19	19
R^2	0.403	0.437

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
OLS and WLS (weighted by sector mean BA, Nov 2022) estimates. Outcome: percentage change in business applications from November 2022 to the most recent month available. Each observation is a 2-digit NAICS sector ($N = 19$).

Table 2: Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) BA Level
Post × AI Exposure	0.204	30.145	-1096.390
	(0.214)	(29.581)	(7366.702)
Observations	4959	4959	4959
R^2	0.982	0.690	0.834
Sector FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by 2-digit NAICS sector.

Table 2 reports the DiD estimates. All three columns pool the entire post-November 2022 period into a single *Post* indicator, with columns (1)–(3) using log business applications, the indexed BA outcome, and raw BA levels respectively.

To translate the coefficient into economically interpretable magnitudes, Table 3 reports the implied percentage difference in business applications relative to a sector at the 25th percentile of AI exposure, evaluated at selected percentiles of the AIOE distribution. A sector at the median of AI exposure is estimated to have had 2.3% more applications than a P25 sector in the post-ChatGPT period; the gap widens to 5.1% at the 75th percentile and

10.7% at the 90th.

Table 3: Implied Effect of AI Exposure on Business Applications

Percentile	AIOE (normalised)	Effect vs. P25 sector (%)
P10	0.220	-2.4%
P25	0.337	+0.0%
P50	0.447	+2.3%
P75	0.585	+5.1%
P90	0.862	+10.7%

Notes: Implied percentage difference in business applications in the post-ChatGPT period relative to a sector at the 25th percentile of normalised AI Industry Exposure (AIOE). Computed as $(\text{AIOE}_p - \text{AIOE}_{P25}) \times \hat{\beta} \times 100$, where $\hat{\beta}$ is the DiD estimate from column (1) of Table 2.

3.3 Event Study

Figure 3 presents the event-study coefficients. The pre-period estimates ($t < 0$) are small and statistically indistinguishable from zero, supporting the parallel-trends assumption.

A handful of coefficients are statistically significant in the first months immediately following the ChatGPT release, suggesting an early, tentative response. These effects then recede into insignificance across roughly a year, before re-emerging persistently from $t = +17$ (April 2024, $p < 0.10$) and $t = +18$ (May 2024, $p < 0.05$) onward. From May 2024, every subsequent monthly coefficient is statistically significant at the 5 % level or better, and point estimates grow monotonically through the end of the sample, reaching 0.48 at $t = +36$.

4 Robustness Checks

4.1 Alternative Business Applications Metrics

We probe robustness by substituting the outcome with the full suite of available BFS series spanning the formation funnel.

On the applications side, we use corrected business applications (CBA), which purge filings unlikely to yield an employer business; high-propensity applications (HBA), which restrict to applicants with a high predicted probability of hiring; and applications with planned wages (WBA), indicating explicit intent to hire.

On the formations side, we use seasonally-adjusted formations within four and eight quarters (SBF4Q, SBF8Q). The realised formation counts (BF4Q, BF8Q) are excluded as the Census has not yet published these beyond 2022 due to publication lags.

Figure 4 presents the corresponding event-study coefficients for all six series. The pre-period estimates are small and insignificant across every panel, ruling out differential pre-trends. Post-launch dynamics are qualitatively consistent: coefficients drift upward at similar timing and reach statistical significance in the same window regardless of which outcome is used.

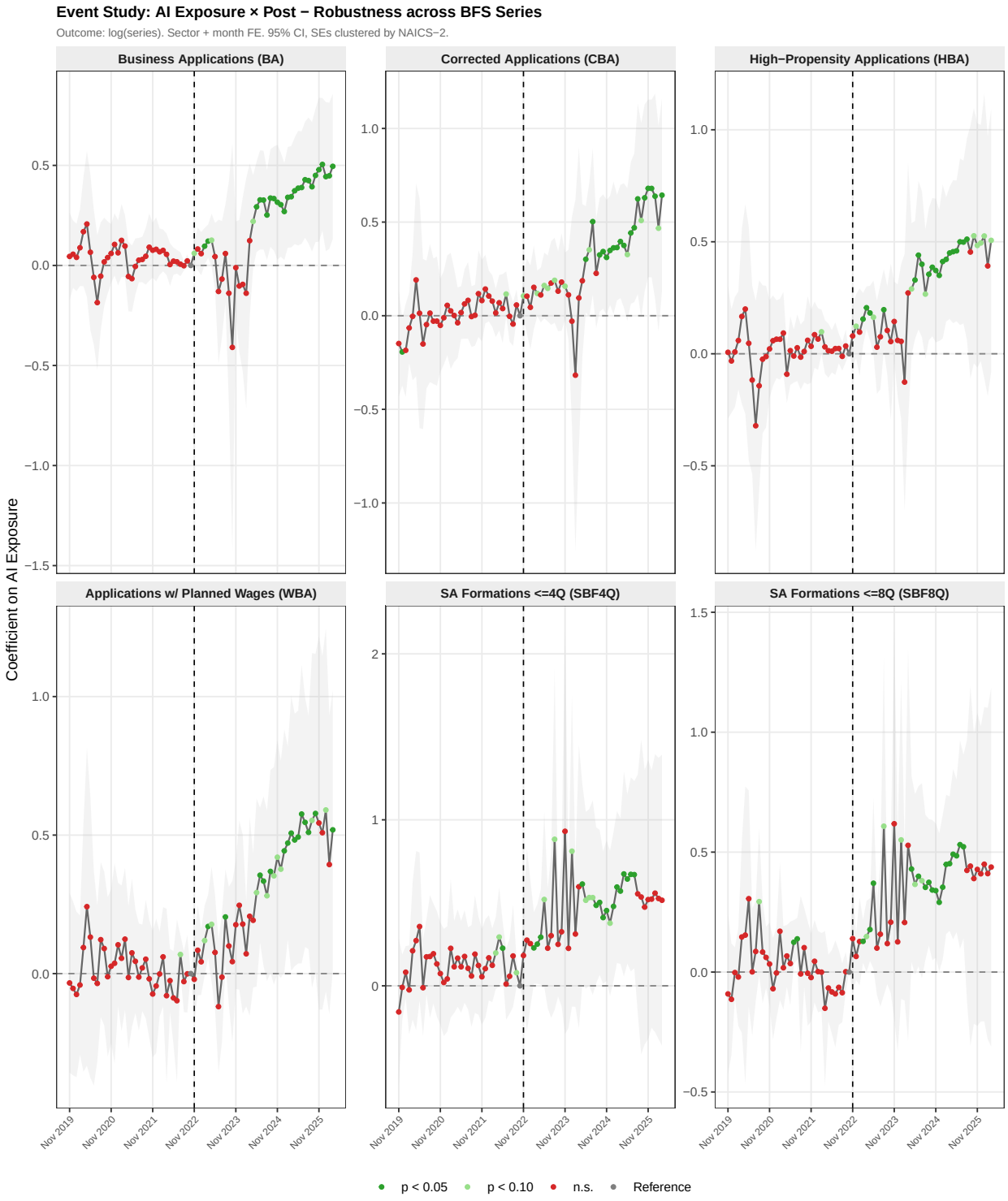

Figure 4: Event-study estimates for six BFS series. Dashed vertical line marks November 2022 ($t = 0$).

Table 4: Robustness: Alternative BFS Series

	(1) log(BA)	(2) log(CBA)	(3) log(HBA)	(4) log(WBA)	(5) log(SBF4Q)	(6) log(SBF8Q)
Post × AI Exposure	0.204	0.438	0.247	0.269	0.594	0.444
	(0.214)	(0.287)	(0.283)	(0.263)	(0.352)	(0.287)
Observations	4959	4959	4959	4959	4951	4744
R^2	0.982	0.980	0.983	0.984	0.983	0.983
Sector FE	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01

Standard errors clustered by 2-digit NAICS sector. All specifications include sector and month fixed effects. BA = total business applications; CBA = corrected business applications; HBA = high-propensity applications; WBA = applications with planned wages; SBF4Q/SBF8Q = seasonally-adjusted formations within 4/8 quarters.

4.2 3-digit NAICS

Figure 5 replicates the motivating scatter at the finer 3-digit NAICS level, comparing mean weekly business applications in November 2022 with November 2025 across 75 sectors. The positive cross-sectional gradient visible at the 2-digit level (Figure 2) persists at higher disaggregation, and the bubble sizes — proportional to pre-period application volume — make clear that larger sectors tend to sit higher on the fitted line. Table **??** formalises this. The unweighted OLS slope is near zero and statistically insignificant ($\hat{\delta} = 0.9$, $p > 0.10$, $R^2 \approx 0.00$), reflecting substantial noise among the many small 3-digit sectors. Once sectors are weighted by their pre-period application volume, however, the slope rises sharply to 41.1 ($p < 0.01$, $R^2 = 0.31$): among the sectors that account for the bulk of economic activity, higher AI exposure is strongly associated with faster application growth.

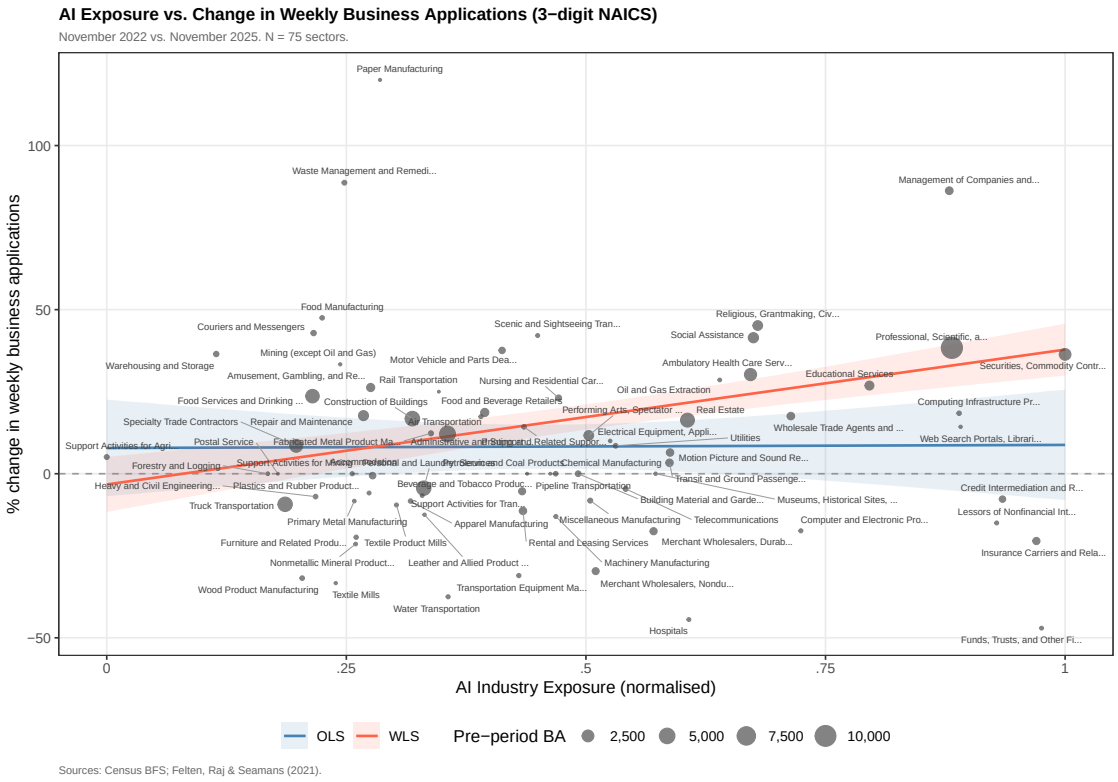

Figure 5: AI exposure versus change in weekly business applications, 3-digit NAICS sectors. November 2022 vs. November 2025. Bubble size proportional to pre-period mean weekly applications. Blue line: unweighted OLS. Red line: WLS weighted by pre-period application volume. Shaded bands: 95 % confidence intervals.

Table 5: Cross-Sectional Regression: AI Exposure vs. Change in Weekly Applications (3-digit NAICS)

	OLS	WLS
AI Exposure (AIOE, normalised)	0.936	41.092***
	(14.332)	(7.223)
Constant	7.840	-3.271
	(7.394)	(4.212)
Sectors (3-digit)	75	75
R^2	0.000	0.307

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
OLS and WLS (weighted by pre-period mean weekly BA) at 3-digit NAICS. Outcome: percentage change in mean weekly business applications from November 2022 to November 2025.

Figure 6 presents the event-study coefficients from estimating equation (2) on the weekly 3-digit NAICS panel. Because the raw weekly series exhibit strong sector-specific seasonality — high-AIOE sectors such as Professional Services have a markedly different within-year cycle than low-AIOE sectors such as Construction — each sector's weekly BA series is first seasonally adjusted using STL decomposition on the log scale (multiplicative, periodic window) before entering the regression. The event study uses 4-week bins, sector and ISO year-week fixed effects, and observations weighted by each sector's pre-period mean weekly BA. Weighting by sector size is motivated by the cross-sectional evidence in Table **??**: the relationship between AI exposure and application growth is concentrated among larger sectors, so weighting ensures that identification comes from the same margin. The pre-period coefficients are small and statistically indistinguishable from zero, supporting the parallel-trends assumption at finer sectoral disaggregation. Post-launch dynamics mirror those in the 2-digit baseline: coefficients drift upward and eventually reach statistical significance, consistent with AI-exposed sectors experiencing a relative surge in weekly business applications following the ChatGPT release.

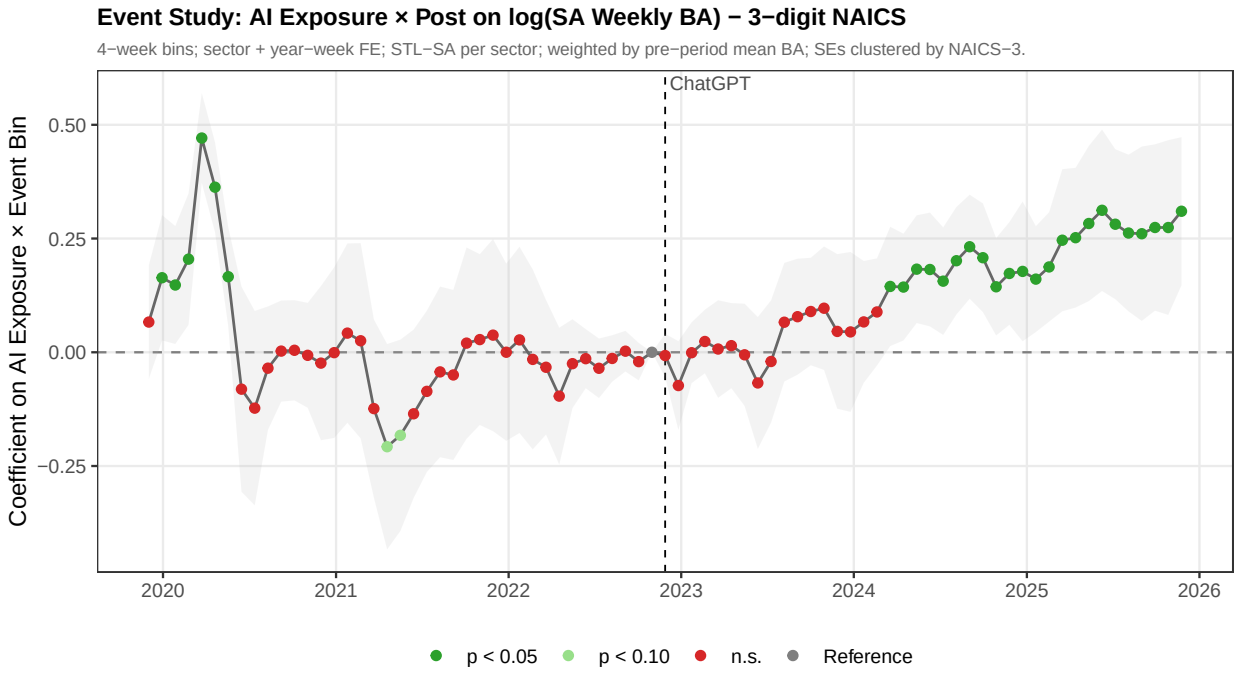

Figure 6: Event-study estimates at 3-digit NAICS. Outcome: log seasonally adjusted weekly business applications (STL decomposition, multiplicative, per sector). 4-week event-time blocks relative to ChatGPT launch (28 Nov 2022). Reference block: $t \in [-4, -1]$. Observations weighted by pre-period mean weekly BA. Sector and ISO year-week fixed effects; SEs clustered by NAICS-3.

4.3 US States

Appendix B reports a state-level robustness check using the AI Geographic Exposure (AIGE) index of Felten et al. (2021) and state × month panels. The state-level estimates are statistically insignificant, likely because state aggregation dilutes the within-sector AI variation that drives the main results; the sector-level estimates remain the primary evidence.

5 Conclusion

[Conclusion placeholder.]

References

Aaron Chatterji, Daniel Rock, and Eduard Talamàs. Transformative AI and firms. In Ajay K. Agrawal, Erik Brynjolfsson, and Anton Korinek, editors, *The Economics of Transformative AI*, chapter 7. University of Chicago Press, 2025. URL https://www.nber.org/chapters/c15327.

Ryan A. Decker and John Haltiwanger. Surging business formation in the pandemic: Causes and consequences? *Brookings Papers on Economic Activity*, Fall:1–74, 2023. URL https://www.brookings.edu/wp-content/uploads/2023/09/Decker-Haltiwanger_16820-BPEA-FA23_WEB.pdf.

Edward W. Felten, Manav Raj, and Robert Seamans. Occupational, industry, and geographic exposure to artificial intelligence: A novel dataset and its potential uses. *Strategic Management Journal*, 42(12):2195–2217, 2021. doi: 10.1002/smj.3325.

A Summary Statistics

Table 6: AI Industry Exposure and Business Formation Statistics by Sector

Sector	NAICS	AIIE	Mean	SD	Min	Max	N
			Monthly BA (thousands)				(months)
Finance & Insurance	52	2.04	14.0	3.6	8.4	22.0	261
Professional, Scientific, Technical Svcs	54	1.71	38.2	11.7	19.8	76.7	261
Management of Companies	55	1.70	2.2	1.3	0.9	6.6	261
Educational Services	61	1.34	4.6	1.5	2.4	8.6	261
Information	51	1.11	6.1	1.8	3.0	11.4	261
Health Care & Social Assistance	62	0.59	18.3	5.6	10.2	33.1	261
Real Estate & Rental	53	0.35	16.9	5.4	8.9	26.5	261
Wholesale Trade	42	0.24	8.9	1.3	6.1	16.2	261
Utilities	22	0.20	0.3	0.2	0.1	0.8	261
Retail Trade (NAICS 44–45)	RET	-0.05	45.2	24.7	20.1	131.0	261
Other Services	81	-0.25	26.6	10.0	12.7	62.8	261
Manufacturing (NAICS 31–33)	MNF	-0.43	5.7	1.3	3.7	10.5	261
Admin. & Support Services	56	-0.58	20.3	7.2	10.9	41.9	261
Arts, Entertainment, & Recreation	71	-0.61	9.5	2.2	5.3	15.4	261
Mining, Quarrying, Oil & Gas	21	-0.66	0.6	0.1	0.3	1.4	261
Transportation & Warehousing (48–49)	TW	-0.95	18.8	9.7	7.9	44.0	261
Construction	23	-1.10	29.7	9.1	16.0	52.9	261
Accommodation & Food Services	72	-1.12	15.6	5.5	8.5	31.6	261
Agriculture, Forestry, Fishing	11	-2.00	3.5	1.5	1.9	19.4	261

Notes: AIIE = AI Industry Exposure index from Felten et al. (2021), constructed from O*NET task descriptions and aggregated from 4-digit to 2-digit NAICS by employment-weighted averaging. Mean, SD, Min, and Max refer to monthly seasonally adjusted business applications (thousands) over the full sample period (January 2006 to early 2026). N is the number of monthly observations. Sectors sorted by AIIE score descending.

B State-Level Robustness

Table 7 presents the state-level DiD estimates. The unit of observation is a state-month cell, with the outcome being log total seasonally adjusted business applications aggregated across all sectors within the state. The AI exposure measure is the AIGE index of Felten et al. (2021), which captures the AI exposure of a state's workforce through its occupational mix. State fixed effects absorb permanent cross-state differences in entrepreneurial culture, regulatory environment, and industrial composition; month fixed effects absorb national macro shocks as before. Standard errors are clustered at the state level (56 clusters).

Figure 7 provides raw descriptive evidence at the state level, analogous to Figure **??** for sectors. States are sorted into quartiles by their AIGE index and the cross-state mean of indexed business applications (January 2020 = 100) is plotted for each quartile.

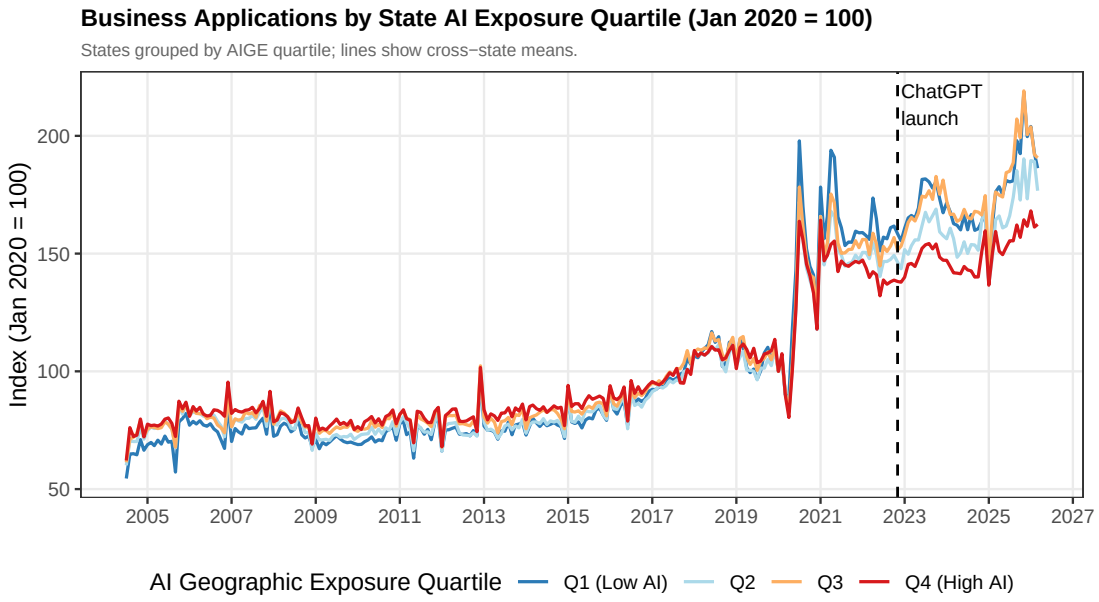

Figure 7: Mean business applications (index, January 2020 = 100) by AI Geographic Exposure (AIGE) quartile across US states and territories. Q1 denotes the lowest-exposure quartile; Q4 the highest. Lines show simple cross-state means within each quartile. Dashed vertical line marks November 2022 (ChatGPT launch).

Column (1) shows the single-period estimate; column (2) uses the indexed BA outcome; column (3) splits the post period as in the main analysis. The state-level estimates are

Table 7: State-Level Difference-in-Differences Estimates

	(1) log(BA)	(2) BA Index	(3) Two-period
$\text{Post}_{\text{ChatGPT}} \times$ AI Exposure	-0.445	-67.191	-0.423
	(0.274)	(42.331)	(0.265)
Observations	13572	13572	13572
R^2	0.986	0.825	0.986
State FE	✓	✓	✓
Month FE	✓	✓	✓

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by state. AI exposure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, constructed from occupation-level AIOE scores weighted by the state employment-mix across occupations. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

statistically insignificant across all specifications, in contrast to the sector-level results in Table 2.

Figure 8 shows the corresponding event-study coefficients, and Figure 9 repeats the exercise across the four BFS series published at the state level (BA, HBA, WBA, CBA; formation series are national-only). Pre-period coefficients are flat across all panels, consistent with parallel pre-trends, but the post-launch coefficients are imprecise and do not reach conventional significance levels in any series.

The absence of a detectable state-level effect likely reflects two compounding limitations. First, states are highly diversified: even a state with a high-AIGE workforce will contain many low-AI sectors that dilute the aggregate application signal. The sector-level design exploits within-sector variation in AI exposure that is averaged away when outcomes are aggregated to the state level. Second, the cross-state dispersion of AIGE is considerably smaller than the cross-sector dispersion of AIOE, reducing statistical power. Together, these features make the state-level panel a demanding test; the sector-level estimates remain the primary evidence.

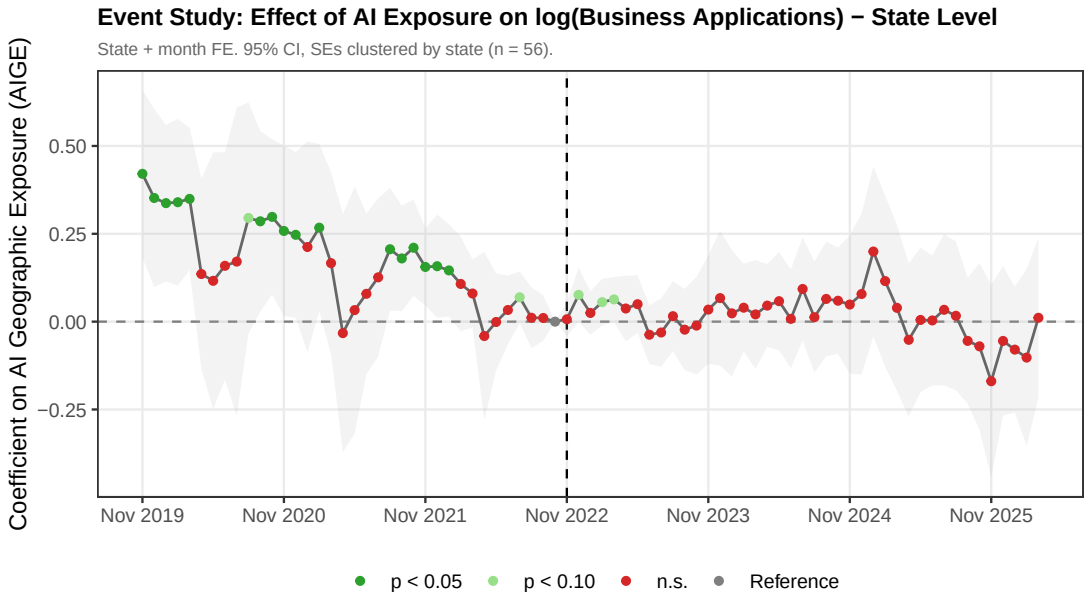

Figure 8: State-level event-study estimates: effect of AI Geographic Exposure (AIGE) on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on state-clustered standard errors (56 clusters). State and month fixed effects included. Coefficients are statistically insignificant throughout.

Figure 9 replicates the state-level event study across the four BFS series available at state level.

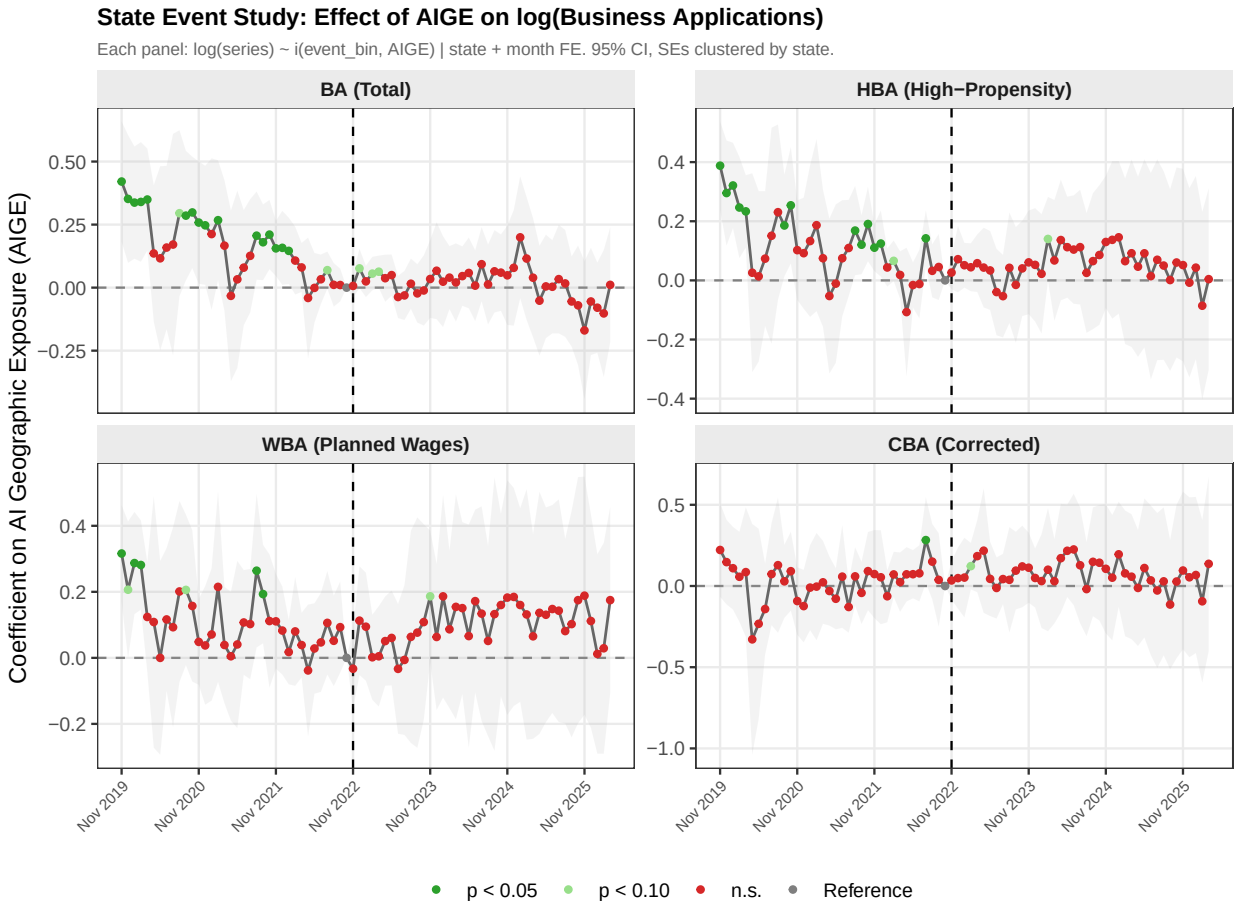

Figure 9: State-level event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI Geographic Exposure (AIGE) and a month-relative-to-November-2022 indicator, estimated on the state × month panel. BA = total applications; HBA = high-propensity applications; WBA = applications with planned wages; CBA = corrected applications. Business formation series (BF4Q, BF8Q, etc.) are not published at state level by the Census Bureau. Shaded bands are 95 % confidence intervals based on state-clustered standard errors (56 clusters). All specifications include state and month fixed effects. Dashed vertical line marks November 2022 ($t = 0$).